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Number of Increasing Paths in a Grid

AMAZON

ADOBE

Problem Statement

Longest increasing path?

The question asks: *How many increasing paths exist?*

Note: A path of length 1 is also a valid path.

Example & Visualization

Consider the following grid:

1	2
3	4

If **Aadya starts at 1**, the possible increasing paths are:

1

1 → 2

1 → 3

1 → 2 → 4

1 → 3 → 4

Similarly, you can trace the paths starting from 2, 3, and 4.

C++ Implementation

```
class Solution {
public:
    const int MOD=1e9+7;
    vector<vector<int>> dp;
    int n;
    int m;
    vector<pair<int,int>> dir={{-1,0},{0,-1},{1,0},{0,1}};
    int dfs(int r,int c,vector<vector<int>>& matrix){

        if(r<0 || c<0 || r>=n || c>=m) return 0;
        if(dp[r][c]!=-1) return dp[r][c]; //already computed

        long long paths=1; //current cell itsself is a valid path

        for(auto &it:dir){
            int nr=r+it.first;
            int nc=c+it.second;

            if(nr>=0 && nc>=0 && nr<n && nc<m){

                if(matrix[nr][nc]>matrix[r][c]){
                    paths+=dfs(nr,nc,matrix);
                    paths%=MOD;

                    //har direction me jaake dekhenge max kaha se mil
rha
                }

            }
        }

        dp[r][c]=paths;
        return dp[r][c];

    }
    int countPaths(vector<vector<int>>& matrix) {
        n=matrix.size();
        m=matrix[0].size();
        long long ans=0;
        dp.assign(n,vector<int>(m,-1));
    }
};
```

```
for(int i=0;i<n;i++){  
    for(int j=0;j<m;j++){  
  
        ans+=dfs(i,j,matrix); //har ek start position se dfs  
krne ki try kro  
        ans%=MOD;  
  
    }  
}  
  
return ans;  
  
}  
};
```